5th

(OSY)

(K scheme)

(Unit 5) (08 Hours, 10 marks)

File management

5.1) File concepts – Attributes, operations, file types and file system structure.

5.2) Accessing methods – sequential & direct

5.3) File allocation methods – contiguous allocation, linked allocation, Indexed allocation.

5.4) directory structure – single level, TWO level, tree structured directory.

5.1) File concepts – Attributes, operations, file types & file system structure

① what is file ?

– File is a collection of related data.

– File is actually a physical memory area for an OS to store the data but for user it is logical place.

② What is directories ?

→ Directories are again like logical compartments which only stores files / sub directories.

Directories are used to organize the data

in a order or manner.

③ What is program ?
→ Programs are set of instructions given to mic to work.

④ What is data file ?
→ Data file is a sequence of bytes, lines, records.

Data files are numeric character, binary.

⑤ what are the extensions of files ?
→ Following are the types of files with different extensions.

File.ppt
File.pps
File.bak
File.doc
File.C
File.gif
File.xls
File.pdf
File.html
File.txt
File.zip
File.jpg
File.obj
File.mp3

⑥ what are the attributes of files ?
→ Files attributes varies from one os to other os.

Following are the attributes of files.

Names → Name is an attribute specified by user.

Identifier → File system gives an unique number that identifies file.

Type → type of file specifies the kind / type of file.

location → It is a place which is specified for user to indicate which file is stored.

size → This attribute is useful to know the size of files in bytes or kb.

protection → This attribute is related with the controls, to specify the users who can do read / write / execute.

time, date & user identification → These attributes are used for the pempose of protection, security.

password → It is an attribute which is secrete code known to system & same to the user of the file.

creator / owner → A creator is a user who has created that file. owner is a user who has owns data files.

max size → Given no of bytes to which file can maximum grow

(7) Explain basic operations on file.

→ Following are the basic operations on file:-

creating a file → To create a file there should be space in the file system for file. Entry of new file must be done in directory.

writing a file → while doing write operation on a file, system call is made which tells both name of file & data to be written in the file. Pointer is a logical reference used when directory entry is done.

Reading a file → It reads data from file. system call states name of file & location of next block of file. once block read then pointer is updated. There are 2 pointer read pointer & write pointer.

Rewind a file → It is also called as operation of repositioning.

Delete a file → To delete a file, we search the directory for the named file.

close a file → when the use of file is over, the file could be closed till the next usage.

Truncate file → Truncating file is the best option instead of deleting that file. It allows users to keep file with attributes as it is.

Append → It is used to add the records of the files at the end of the file. It generally used in database files. This is write type operation.

Seek → It searches data in file depending on different conditions provided by the user.

Get attributes → It is necessary for system to get the attributes of files. For which get attribute system call can be used.

Set attributes → It is a system call to set certain properties of files like hidden, archieve.

Rename → This operation to give new name to file.

(8) List different types of files & explain:

Following are the types of files.

- Simple (Regular file)

File which contain user information are called as regular files / simple files.

- Basic files (directories) :- Files are used for maintaining the structure of file system. They are also called as system files.

- character special files.

(logical files) :- These files are related to printers.

(Block special files) :- (physical files) :- These files are related to disk.

- simple files are also called as binary files. sequential files are related to magnetic tape. called as physical files.

(9) Explain File structure:

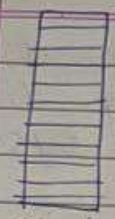
→ Files can be structured by following ways :—

- unstructured sequence of bytes (byte sequence)

- sequence of fixed length of records (Record sequence)

- Time tree / tree records of same / different length.

(tree sequence)



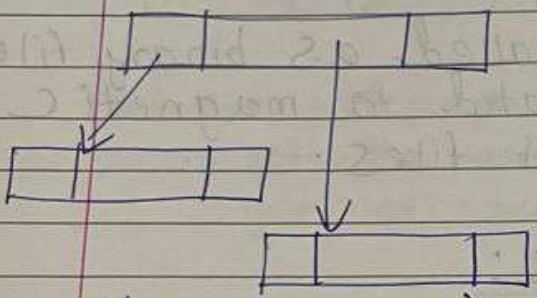
In this file structure, contents are stored in the form of bytes.

(byte sequence)



In this file structure, data is stored in the form of records. Records are collection of related data.

(record sequence)



In this file structure, data is stored in the form of tree.

(tree sequence)

(5.12) Accessing methods – sequential & direct.

→ Following are the file access methods –

- sequential access method

· Most imp operations are read & write.

· Read operation read the file & read pointer go ahead.

· In write operation, write pointer

go to the end of the file & then contents added by the user.

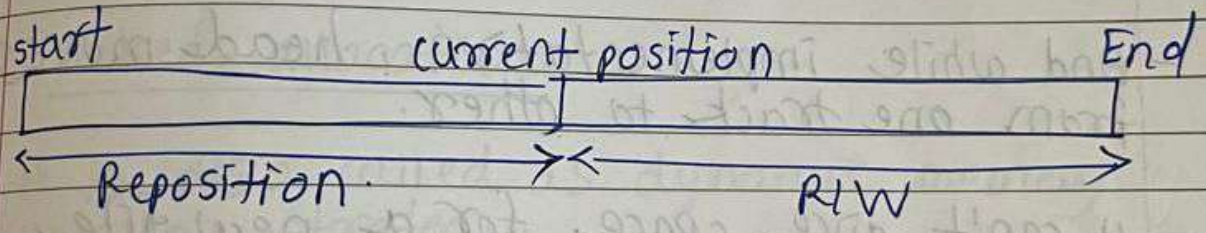
· File access sequentially is a sequential access method, which is tape model of a file.

- direct access method

· An alternative access method to the sequential access is called as direct access method.

· It is based on disk model of a file.

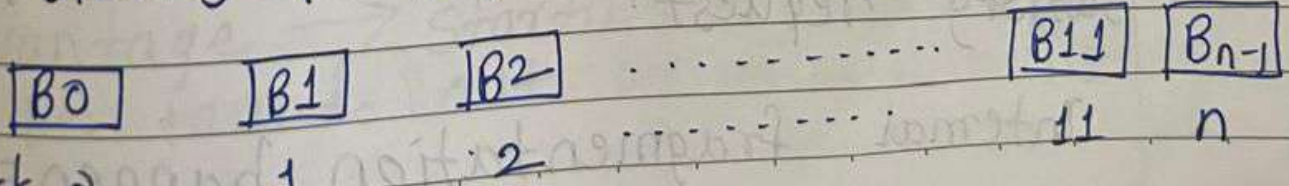
· In this method, file can be viewed as a sequence of blocks that are numbered.



· Block is fixed length unit by OS.

· The major aspect of a direct access method is to allow reading / writing blocks.

· It doesn't keep restriction on the order of blocks & sequence of operations.
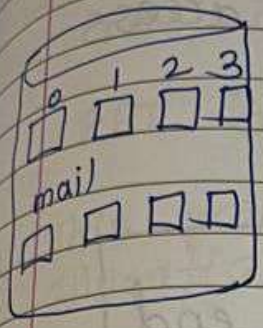


(5.3) File allocation methods

- contiguous method
- linked allocation method
- Indexed allocation method.

① contiguous allocation method

- In this memory allocation method for files & directories, layout of file is in contiguous disk blocks.
- It is used in IBM interactive systems.
- Also in this method, files occupies a set of address on the disk contiguous block.
- And while implementation, head moves from one track to other.
- It can't give space for a new file.
- It suffers because of fragmentation.
- Fragmentation is a process of breaking down the memory space in small pieces.
- Extemal fragmentation is the process which happens when there is total enough memory space available to satisfy request.
- Intemal fragmentation happens when

process needs a memory.



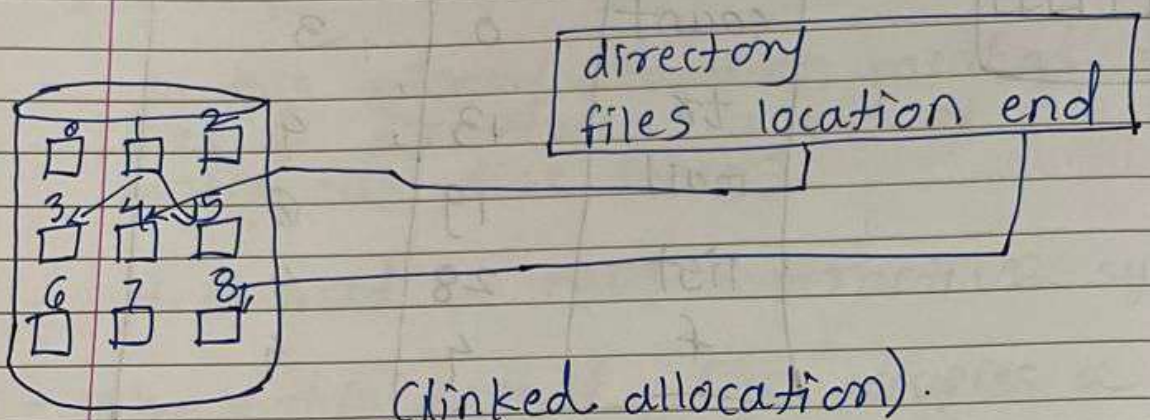
directory		
file	start	length
count	0	3
tr	13	4
mail	19	6
list	28	4
f	4	4

② Linked allocation.

- It is also called as dynamic memory allocation strategy.
- Here contiguous allocation problem solved.
- Here each file is a like list of disk blocks. The directory contain pointer to the 1st & last block of file. Each pointer points to the block & directory / file get allocated to disk block.
- Advantage → starting & ending of file can be found out by the link list.

-disadvantage → pointers needs additional memory space.

It can't used for direct access files.



(linked allocation).

linked allocation	contiguous allocation
- used for sequential access files.	- used for direct access file structure
- not suffer because of external fragmentation.	- it suffers due to fragmentation.
- pointers needs space.	- pointers not required.
File can grow to it's size	- problem when file grow.

③ Indexed allocation

- linked allocation can't perform well if FAT is not there.

FAT is for directory entries.

This problem is recovered by indexed allocation.

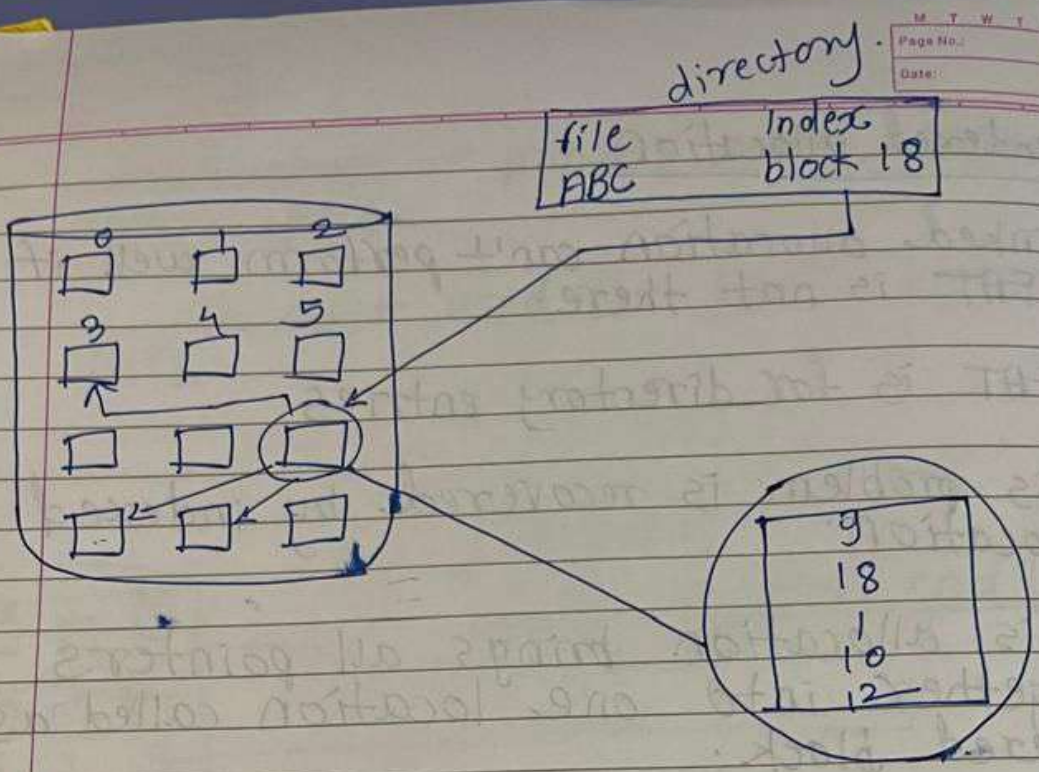
This allocation brings all pointers together into one location called as indexed block.

If we use 1st block of file, then 1st address of index block entry get considered.

- Disadvantage :- space wastage for indexed blocks.

- Advantage :- It is faster than contiguous & linked allocation. Access Index block is easy.

It not suffers from fragmentation.

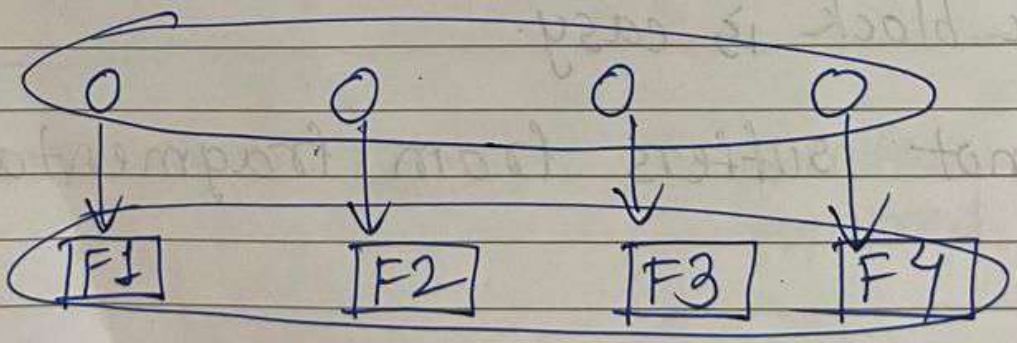


(5.4) Directory structure :— single level, two level, structured directory.

→ (1) directory is logical storage unit where files can be stored.

(2) directory structure is a collection of nodes containing information about all files.

(3)



(4) It is also called as logical structure

(5) physical structure is related to physical devices that stores data.

logical structure is related to logical unit that stores data.

(6) Following are the general directory structure.

- single level directory.
- two level directory
- tree structured directories.

(7) In single level directory, all files are present in single / same directory. This is easy to understand.

When no of files are increasing in single directory then name should be unique.

For the user, it is difficult to remember names of files from single directory. There is a problem with naming & grouping.

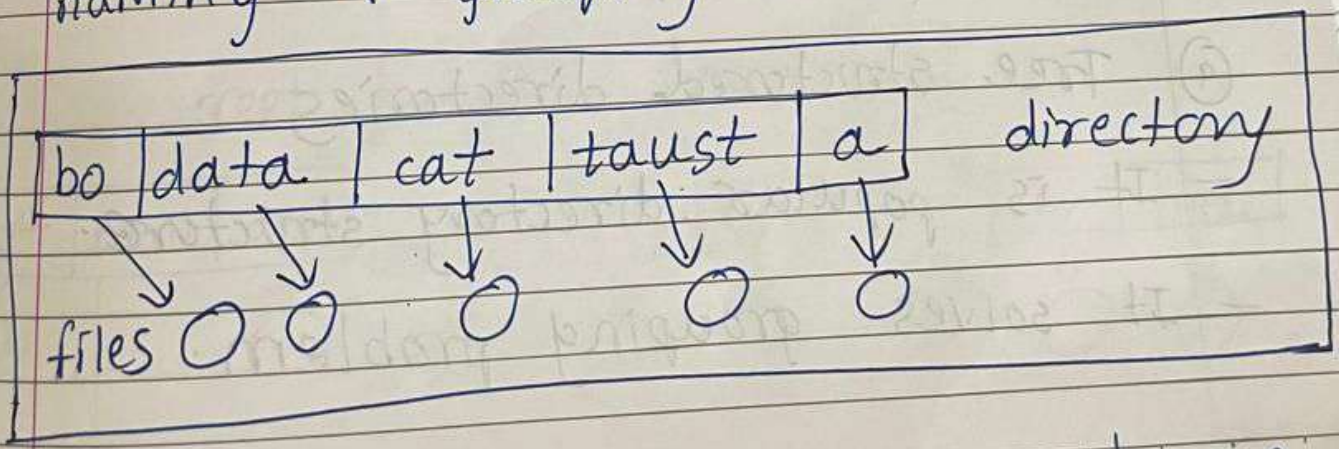


Figure shows files in circular shapes,

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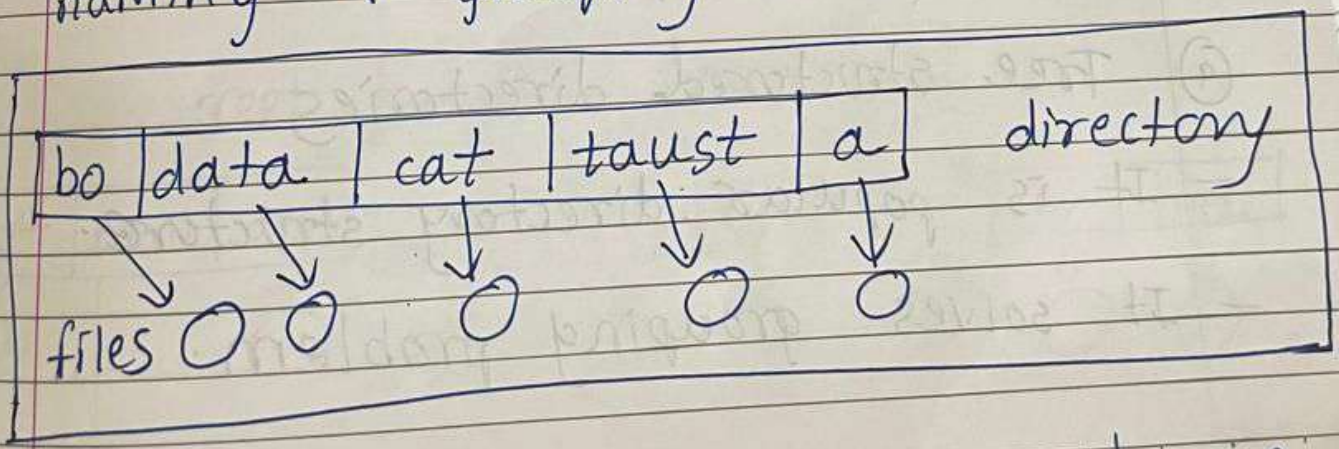


Figure shows files in circular shapes,

- data can be searched due to path. path for file / directory.

- It allows multiple users to share multiple files, directories.

- directories can contain sub directories.

- It helps user to organize data in manner.

- There are 2 paths in tree to search files / directories.

- Relative path starts with current working directory.

Absolute path starts when file is not in same directory.

- user can't delete root directory. os allows to delete sub directories first. when root directory becomes empty then user can delete it.

